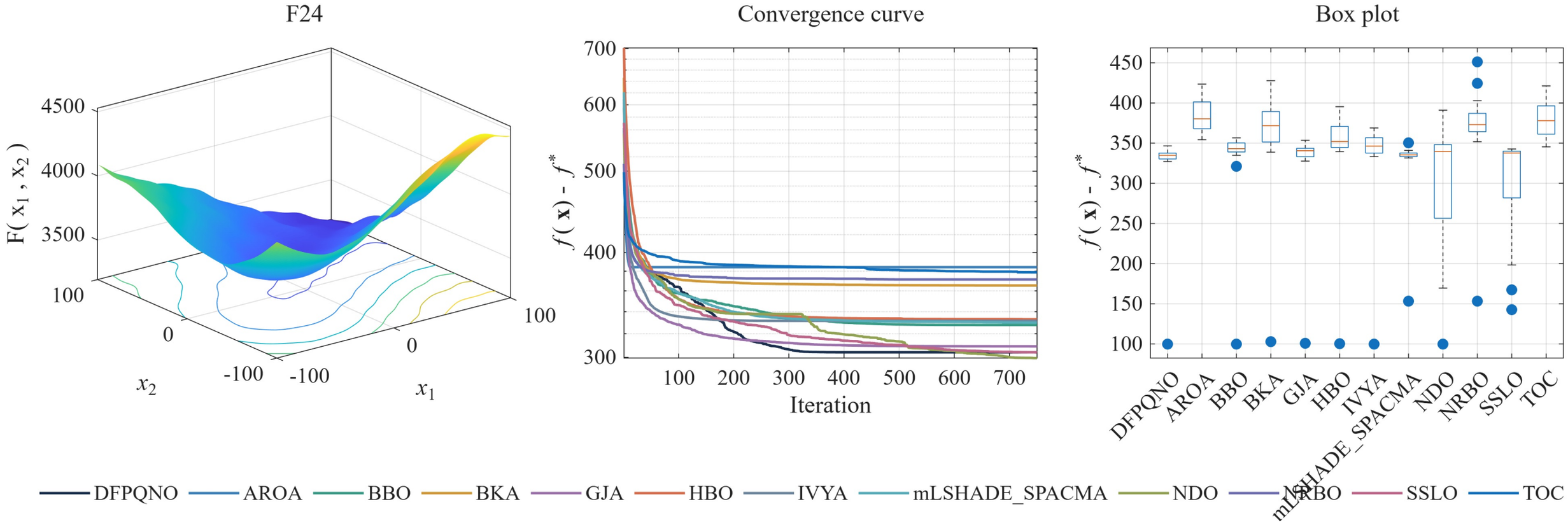

F24
F(x1 , x2)
4500
4000
3500
100
0
-100
x2
-100
0
100
x1
Convergence curve
f(x) - f*
700
600
500
400
300
100
200
300
400
500
600
700
Iteration
Box plot
f(x) - f*
450
400
350
300
250
200
150
100
DFPQNO
AROA
BBO
BKA
GJA
HBO
IVYA
mLSHADE_SPACMA
NDO
NRBO
SSLO
TOC
DFPQNO
AROA
BBO
BKA
GJA
HBO
IVYA
mLSHADE_SPACMA
NDO
NRBO
SSLO
TOC